Acondicionamiento analógico de señales

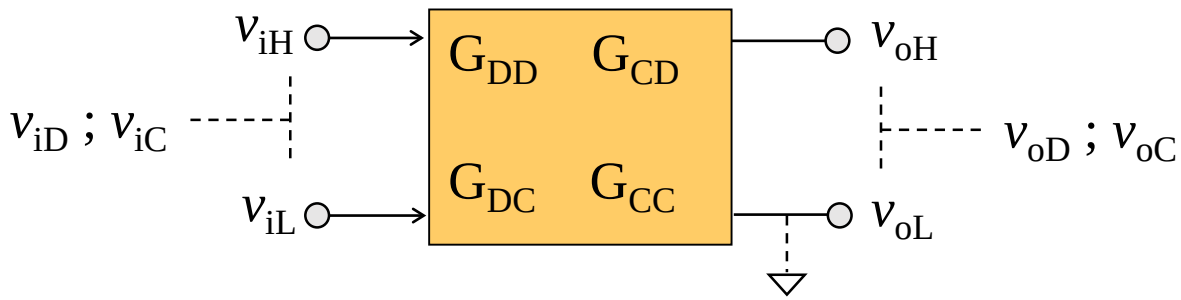
CMRR y A.I.

I&CI: ACONDICIONAMIENTO – CMRR Y A.I.

Lista de temas:

1) Definición CMRR
2) Amplificador Diferencial
3) Amplificador de Instrumentación de 2 AO
4) Amplificador de Instrumentación Full Diferencial
5) CMRR multietapa
6) Amplificador de Instrumentación de 3 AO

1) Definición CMRR



$$v_{iD} = v_{iH} - v_{iL}$$

$$v_{iC} = \frac{v_{iH} + v_{iL}}{2}$$

$$\begin{bmatrix} v_{iD} \\ v_{iC} \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} v_{iH} \\ v_{iL} \end{bmatrix}$$

Rechazo de Modo Común (CMRR)

$$CMRR = \frac{G_{DD}}{G_{DC}}$$

1) Definición CMRR

Muchas no-idealidades de los amplificadores dependen de los AO elegidos.

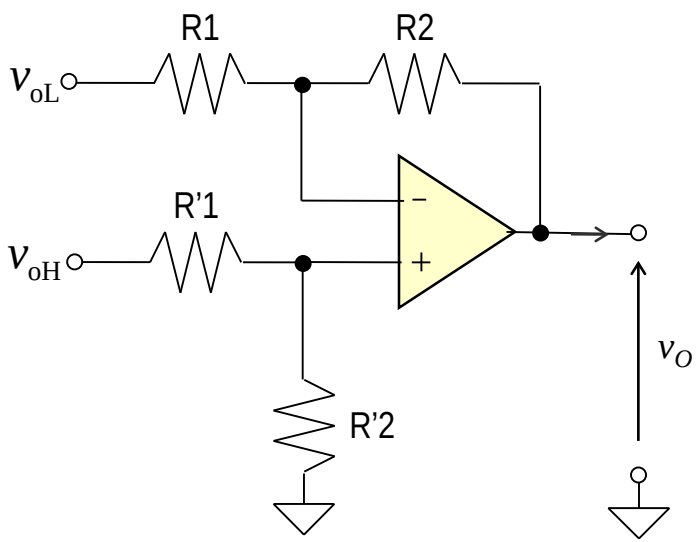
Pero otras, como el Rechazo de Modo Común (CMRR), dependen también de la topología.

A continuación analizamos el CMRR para distintas topologías de amplificación:

2) Amplificador Diferencial

$$v_o = G_{DD} \cdot v_{iD} + G_{DC} \cdot v_{iC}$$

Ganancia diferencial:

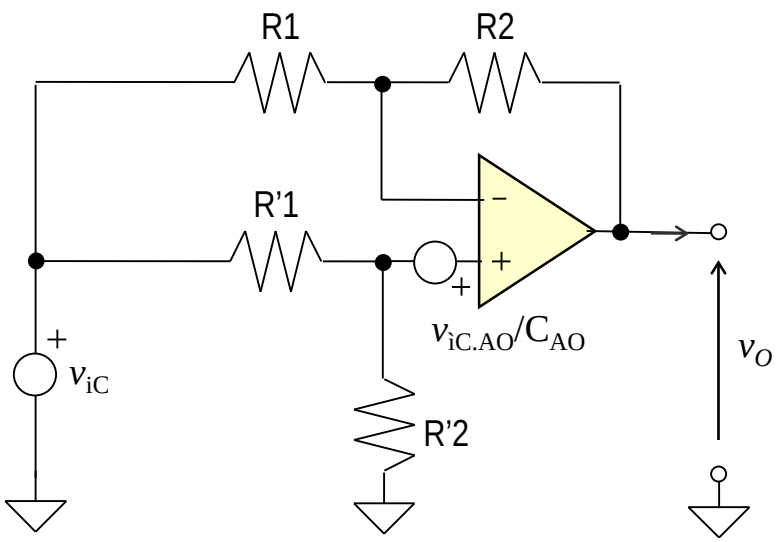


$$v_o = v_{oH} \frac{R_2'}{R_1' + R_2'} \left(1 + R_2 / R_1\right) - v_{oL} \frac{R_2}{R_1}$$

Si $\frac{R_2}{R_1} = \frac{R_2'}{R_1'}$: $\quad v_o = \left(v_{oH} - v_{oL}\right) \frac{R_2}{R_1}$

$$G_{DD} = R_2 / R_1$$

2) Amplificador Diferencial - Efecto del CMRR propio del AO:



$$v_o = \frac{v_{iC.AO}}{C_{AO}} \cdot \left\{ 1 + \frac{R_2}{R_1} \right\}$$

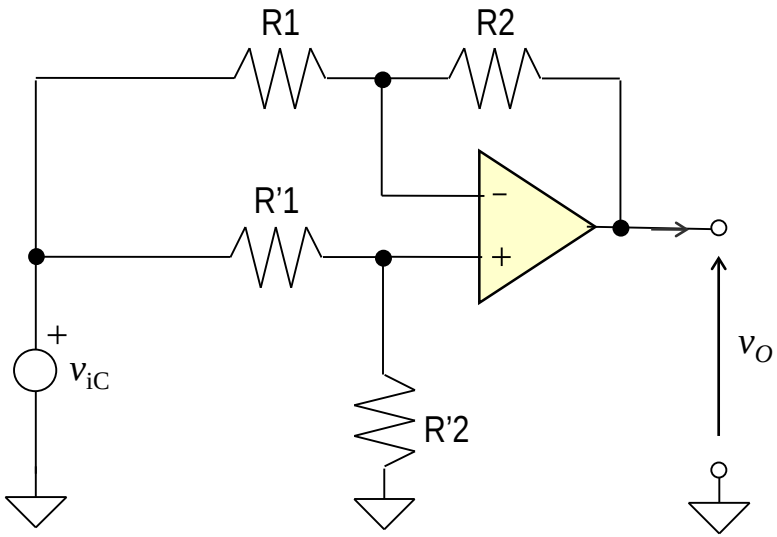
$$v_{iC.AO} = v_{iC} \cdot \frac{R'_2}{R'_1 + R'_2}$$

$$v_o = \frac{v_{iC}}{C_{AO}} \cdot \frac{R_2}{R_1} \frac{1 + R_1/R_2}{1 + R'_1/R'_2}$$

Asumiendo Gdd>>1:

$$G_{DC}\Big|_{AO} \cong \frac{G_{DD}}{C_{AO}}$$

2) Amplificador Diferencial - Efecto del desbalance de los resistores:



Haciendo superposición de la amplificación inversora y no inversora, y agrupando:

$$G_{DC}\big|_R = G_{DD}\left\{\frac{R_1/R_2 - R_1'/R_2'}{1+R_1'/R_2'}\right\}$$

Considerando peor caso (t=$\Delta R/R$): $R_1 = R_{10}(1+t)\ ;\ R_2 = R_{20}(1-t)\ ;\ R_1' = R_{10}(1-t)\ ;\ R_2' = R_{20}(1+t)$

$$G_{DC}\big|_R = G_{DD}\frac{\dfrac{R_{10}(1+t)}{R_{20}(1-t)} - \dfrac{R_{10}(1-t)}{R_{20}(1+t)}}{1+\dfrac{R_{10}(1-t)}{R_{20}(1+t)}} \cong G_{DD}\frac{\dfrac{R_{10}(1+t)}{R_{20}(1-t)} - \dfrac{R_{10}(1-t)}{R_{20}(1+t)}}{1+\dfrac{R_{10}}{R_{20}}} \Rightarrow \boxed{G_{DC}\big|_R \cong G_{DD}\frac{4t}{1+G_{DD}}}$$

2) Amplificador Diferencial

$$G_{DC}\big|_{AO} \cong \frac{G_{DD}}{C_{AO}}$$

$$G_{DC}\big|_{R} \cong G_{DD}\frac{4t}{1+G_{DD}}$$

$$G_{DC.T} = G_{DC}\big|_{AO} + G_{DC}\big|_{R} = G_{DD}\left(\frac{4t}{1+G_{DD}} + \frac{1}{C_{AO}}\right)$$

$$CMRR^{-1} = \frac{4t}{1+G_{DD}} + \frac{1}{C_{AO}}$$

Para que ambos términos sean comparables:

$$t \approx \frac{1+G_{DD}}{4C_{AO}}$$

Considerando un A.O. con un CMRR de unos 100dB (100.000 veces), y una G_{DD} de 100 veces, resulta una tolerancia de 0.025%!!!

Luego, en gral.:

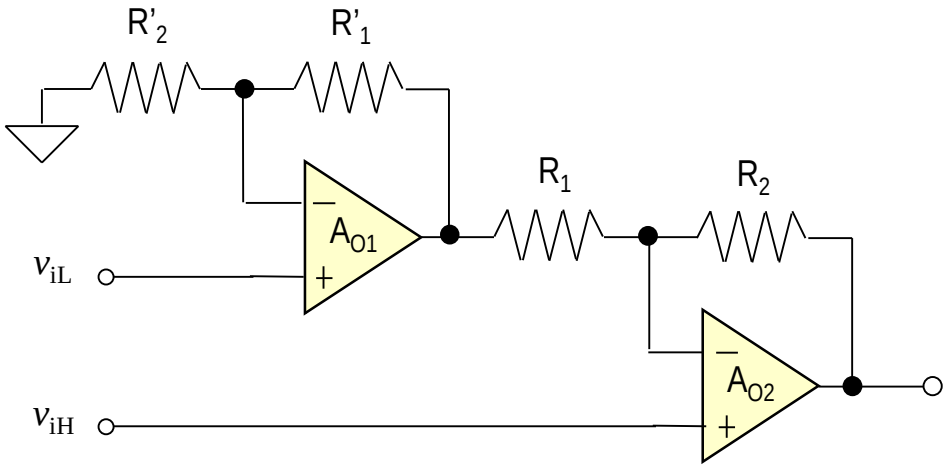
$$CMRR \cong \frac{1+G_{DD}}{4\cdot t}$$

2) Amplificador Diferencial

Algunas Conclusiones:

- Adaptador de terminales.
- Para mejorar CMRR: Gdd alta (no es la aplicación típica)
  o
  resistores de muy alta precisión (más caros)
- Baja Zin…

3) Amplificador de Instrumentación de 2 AO



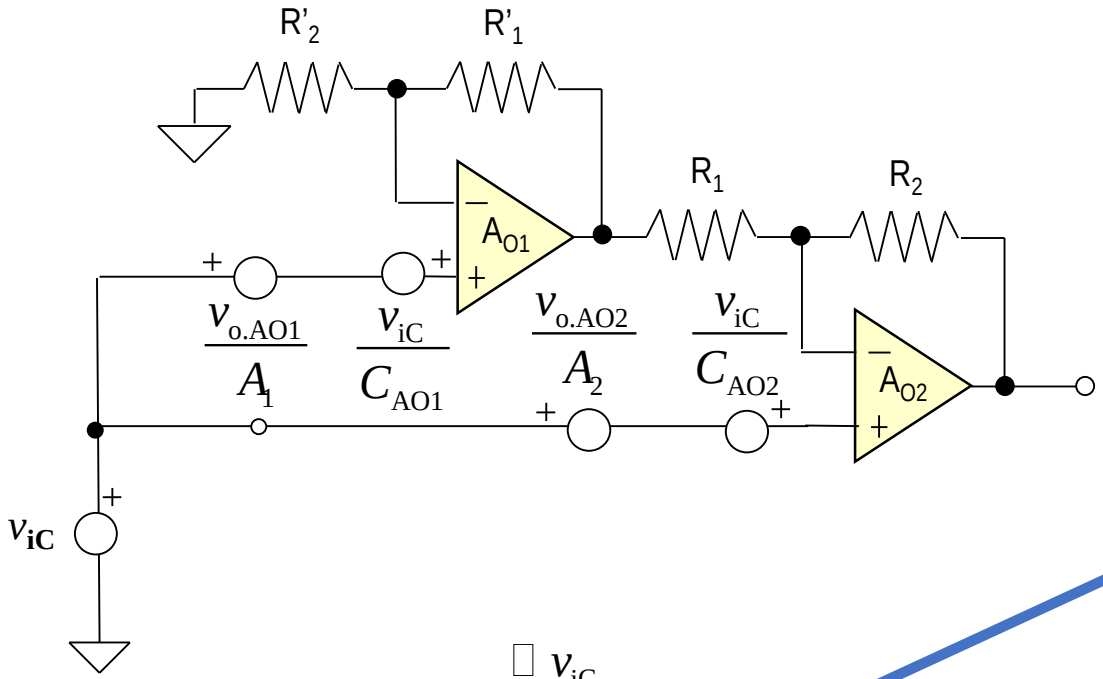
$$G_{DD} = ??$$

$$v_o = v_{iH}\left(1 + R_2/R_1\right) - v_{iL}\left(1 + R_1'/R_2'\right)\left(R_2/R_1\right)$$

$$\text{Si} \quad \underbrace{R_2 = R_2' \quad ; \quad R_1 = R_1'}_{\text{Condición necesaria para amp. diferencia!}} : \quad v_o = \left(v_{iH} - v_{iL}\right)\left(1 + R_2/R_1\right) \Rightarrow G_{DD} = 1 + R_2 / R_1$$

3) Amplificador de Instrumentación de 2 AO

En este caso se suma el efecto de la ganancia finita de los AO:



Asumiendo Gdd suf. grande:

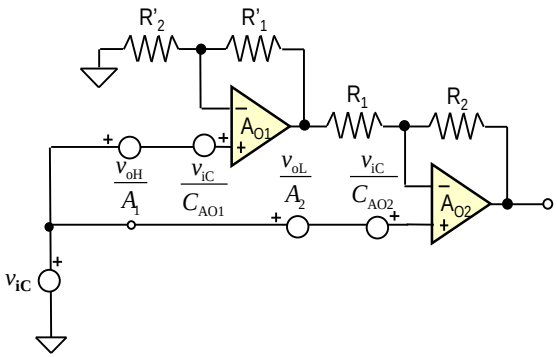
$$v_{o.AO1} \cong v_{iC}\left(1+\frac{R_1'}{R_2'}\right) \cong v_{iC}$$

$$v_{o.AO2} \cong \frac{v_{iC}}{CMRR}$$

$$v_{o.AO2} \ll v_{o.AO1}$$

$$v_o \cong \left[v_{iC}\left(C_{AO2}^{-1}-C_{AO1}^{-1}\right)+\left(\underbrace{v_{o.AO1}A_1^{-1}}_{\ll v_{iC}}-\cancel{v_{o.AO2}A_2^{-1}}\right)\right]\left(1+\frac{R_2}{R_1}\right)+v_{iC}\left[\left(1+\frac{R_2}{R_1}\right)-\left(1+\frac{R_1'}{R_2'}\right)\frac{R_2}{R_1}\right]$$

3) Amplificador de Instrumentación de 2 AO



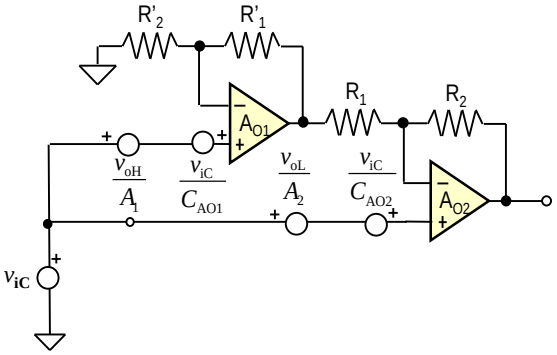
$$v_o \cong v_{iC}\left[\left(C_{AO2}^{-1}-C_{AO1}^{-1}\right)+A_1^{-1}\right]\left(1+\frac{R_2}{R_1}\right)+v_{iC}\left(1-\frac{R_1'}{R_2'}\frac{R_2}{R_1}\right)$$

$$G_{DC} \cong \left[\left(C_{AO2}^{-1}-C_{AO1}^{-1}\right)+A_1^{-1}\right]G_{DD}+\left(1-\frac{R_1'}{R_2'}\frac{R_2}{R_1}\right)$$

Considerando peor caso (hacer cuentas):

$$G_{DC} \cong \left[\left(C_{AO2}^{-1}-C_{AO1}^{-1}\right)+A_1^{-1}\right]G_{DD}+4t \qquad ; \qquad t=\frac{\Delta R}{R}$$

3) Amplificador de Instrumentación de 2 AO



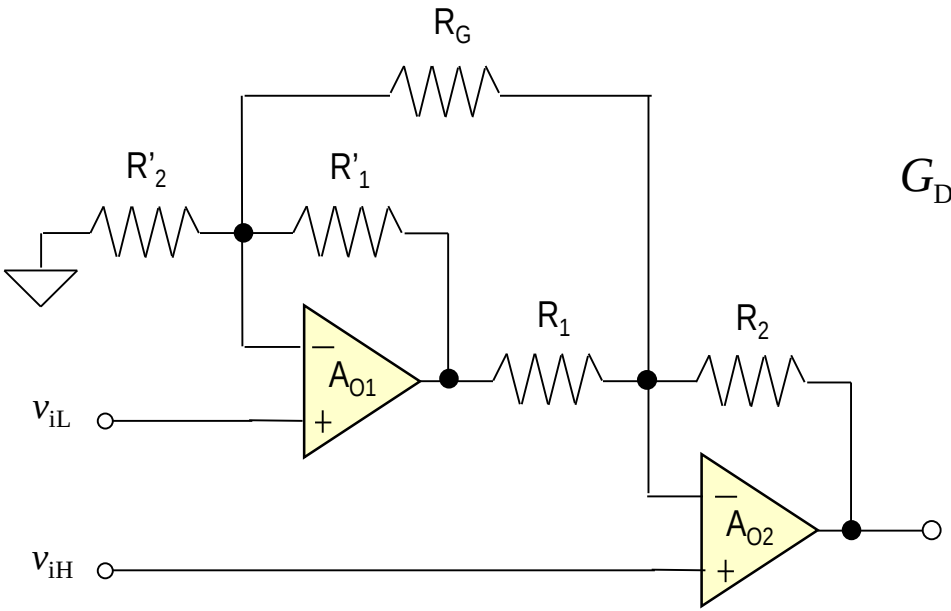
$$CMRR^{-1} \cong \left[\left(C_{\mathrm{AO2}}^{-1} - C_{\mathrm{AO1}}^{-1}\right) + A_1^{-1}\right] + \frac{4t}{G_{\mathrm{DD}}}$$

$$CMRR^{-1} \cong A_1^{-1} + \frac{4t}{G_{\mathrm{DD}}}$$

3) Amplificador de Instrumentación de 2 AO

Contra: Se deben modificar dos resistores para modificar la ganancia.

Alternativa: A. I. de 2 A.O. con ajuste de ganancia mediante un único resistor



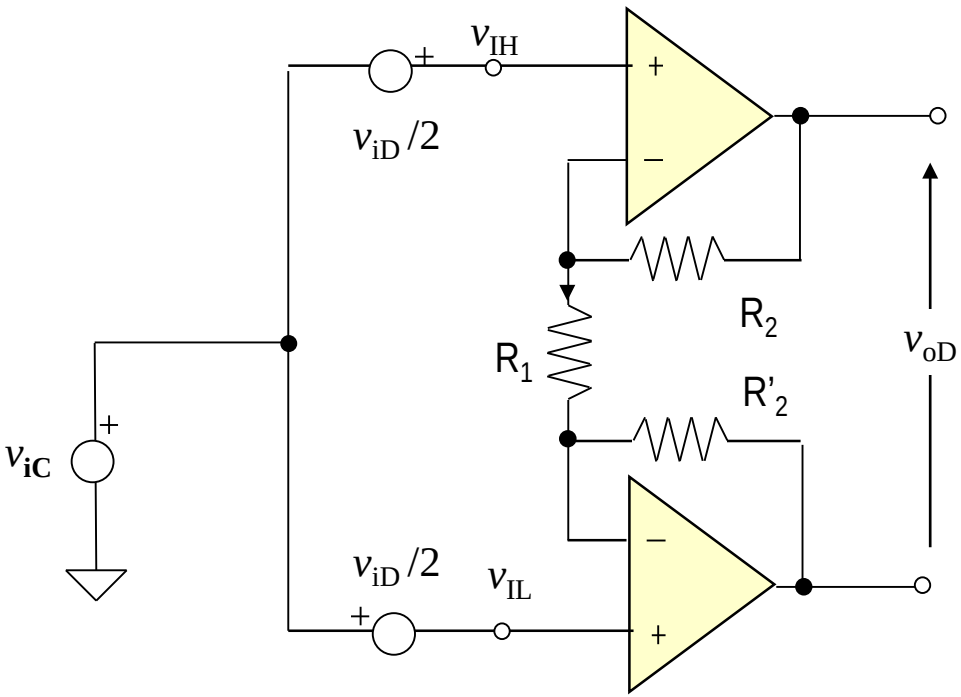
$$G_{DD} = 1 + \frac{R_2}{R_1} + 2\frac{R_2}{R_G}$$

3) Amplificador de Instrumentación de 2 AO

Algunas Conclusiones:

- Adaptador de terminales.
- Alta Zin.
- CMRR sigue dependiendo del desbalance entre resistores!
- Si Gdd grande: $CMRR \cong A_1$, el cual será pobre a altas frecs.

4) Amplificador de Instrumentación Full Diferencial



$$i_{R1} = \frac{v_{iH} - v_{iL}}{R_1}$$

$$v_{oD} = i_{R1} \cdot \left(R_1 + R_2 + R_2' \right)$$

$$v_{oD} = \left(v_{iH} - v_{iL} \right) \cdot \frac{R_1 + R_2 + R_2'}{R_1}$$

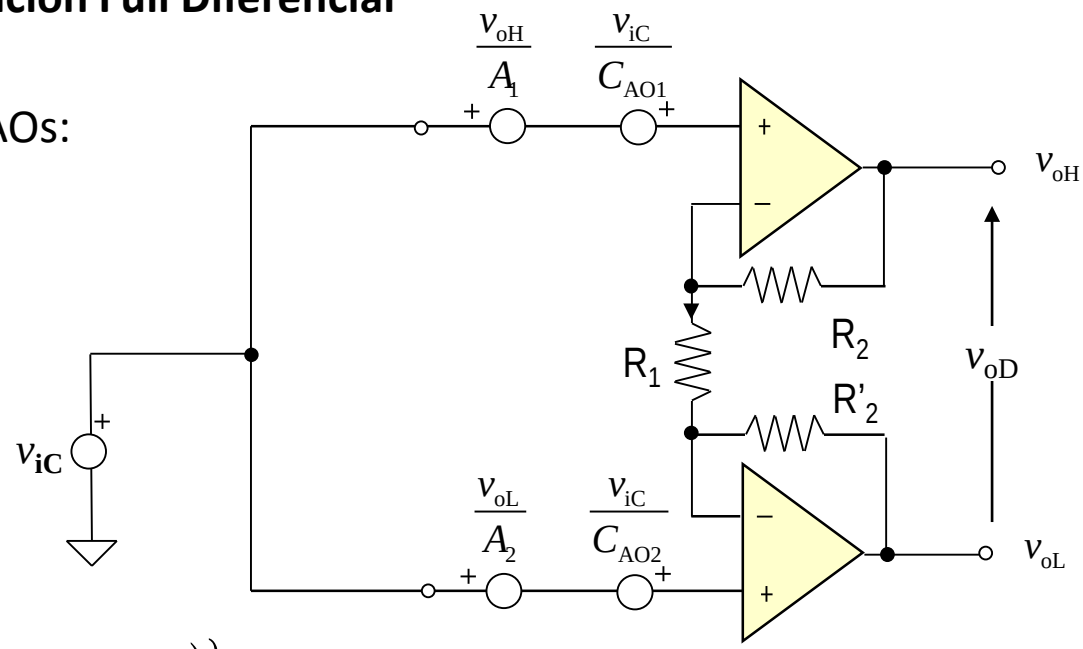
$$v_{oD} = v_{iD} \cdot \left(1 + 2R_2 / R_1 \right)$$

$$G_{DD} = \left(1 + 2R_2 / R_1 \right) \quad ; \quad G_{DC} = 0$$

(por desbalance resistores)

4) Amplificador de Instrumentación Full Diferencial

Considerando no-idealidades AOs:



$$G_{DD} = 1 + 2\,R_2/R_1 \quad ;$$

$$G_{DC} = G_{DD} \cdot \left\{ \left(C_{AO2}^{-1} - C_{AO1}^{-1} \right) + \left(A_1^{-1} - A_2^{-1} \right) \right\};$$

$$G_{CC} = 1$$

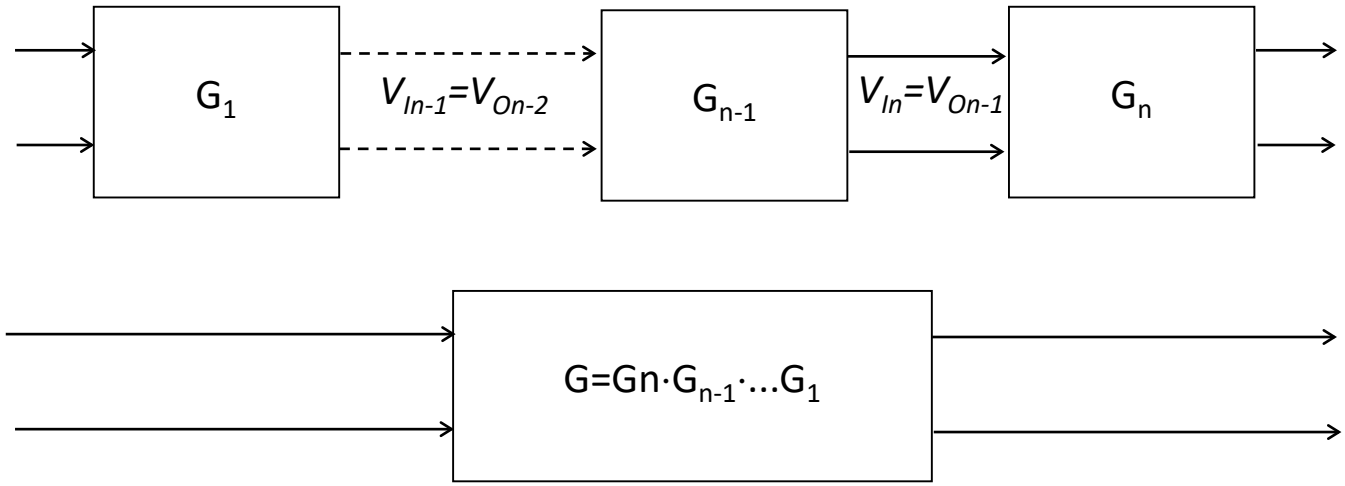
$$CMRR^{-1} = \left(C_{AO2}^{-1} - C_{AO1}^{-1} \right) + \left(A_1^{-1} - A_2^{-1} \right)$$

4) Amplificador de Instrumentación Full Diferencial

Algunas Conclusiones:

- Alta Zin.
- Alto CMRR (no depende de desbalance de resistores).
- Salida diferencial!

5) CMRR multietapa

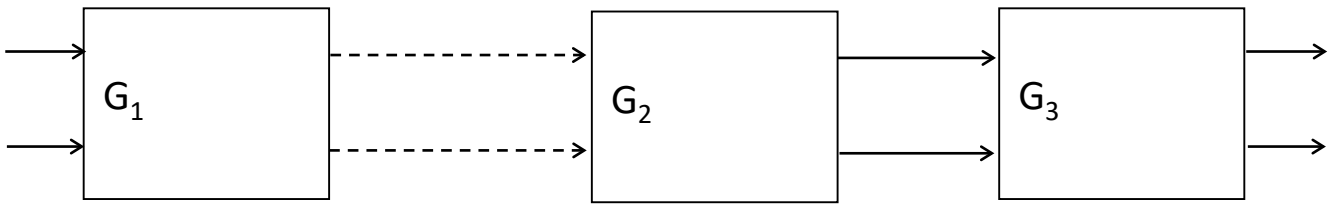


$$G = G_n \cdot G_{n-1} \cdot \cdots \cdot G_1 = \prod_{i=n}^{1} G_i$$

Ahora bien:

$$CMRR = \frac{G_{DD}}{G_{DC}}$$

5) CMRR multietapa



$$v_{\mathrm{oD}} = v_{\mathrm{iC}} G_{\mathrm{DC1}} G_{\mathrm{DD2}} G_{\mathrm{DD3}} + v_{\mathrm{iC}} G_{\mathrm{CC1}} G_{\mathrm{DC2}} G_{\mathrm{DD3}} + v_{\mathrm{iC}} G_{\mathrm{CC1}} G_{\mathrm{CC2}} G_{\mathrm{DC3}} + \cancel{v_{\mathrm{iC}} G_{\mathrm{CC1}} G_{\mathrm{CD2}} G_{\mathrm{DD3}}} + \cdots$$

$$G_{\mathrm{DC.T}} \cong G_{\mathrm{DC1}} G_{\mathrm{DD2}} G_{\mathrm{DD3}} + G_{\mathrm{CC1}} G_{\mathrm{DC2}} G_{\mathrm{DD3}} + G_{\mathrm{CC1}} G_{\mathrm{CC2}} G_{\mathrm{DC3}}$$

$$CMRR_{\mathrm{T}}^{-1} = \frac{G_{\mathrm{DC.T}}}{G_{\mathrm{DD.T}}} \cong \frac{G_{\mathrm{DC1}} G_{\mathrm{DD2}} G_{\mathrm{DD3}} + G_{\mathrm{CC1}} G_{\mathrm{DC2}} G_{\mathrm{DD3}} + G_{\mathrm{CC1}} G_{\mathrm{CC2}} G_{\mathrm{DC3}}}{G_{\mathrm{DD1}} G_{\mathrm{DD2}} G_{\mathrm{DD3}}}$$

5) CMRR multietapa

$$CMRR_{\mathrm{T}}^{-1} \cong \frac{G_{\mathrm{DC1}}}{G_{\mathrm{DD1}}} + \frac{G_{\mathrm{DC2}}}{G_{\mathrm{DD2}}}\frac{G_{\mathrm{CC1}}}{G_{\mathrm{DD1}}} + \frac{G_{\mathrm{DC3}}}{G_{\mathrm{DD3}}}\frac{G_{\mathrm{CC1}}G_{\mathrm{CC2}}}{G_{\mathrm{DD1}}G_{\mathrm{DD2}}}$$

Llamando:

Rechazo de Modo Común (CMRR): $C = \dfrac{G_{DD}}{G_{DC}}$

Factor de discriminación: $D = \dfrac{G_{DD}}{G_{CC}}$

$$CMRR_{\mathrm{T}}^{-1} \cong C_1^{-1} + \left(D_1 C_2\right)^{-1} + \left(D_1 D_2 C_3\right)^{-1}$$

5) CMRR multietapa

$$CMRR_1 = \frac{G_{DD1}}{G_{DC1}} = C_1$$

CMRR segunda etapa

$$CMRR_2 = \frac{G_{DD1}}{G_{CC1}} \cdot \frac{G_{DD2}}{G_{DC2}} = D_1 \cdot C_2$$

=1

$$CMRR_3 = \frac{G_{DD1}}{G_{CC1}} \cdot \frac{G_{DD2}}{G_{CC2}} \cdot \frac{G_{DD3}}{G_{DC3}} = D_1 \cdot D_2 \cdot C_3$$

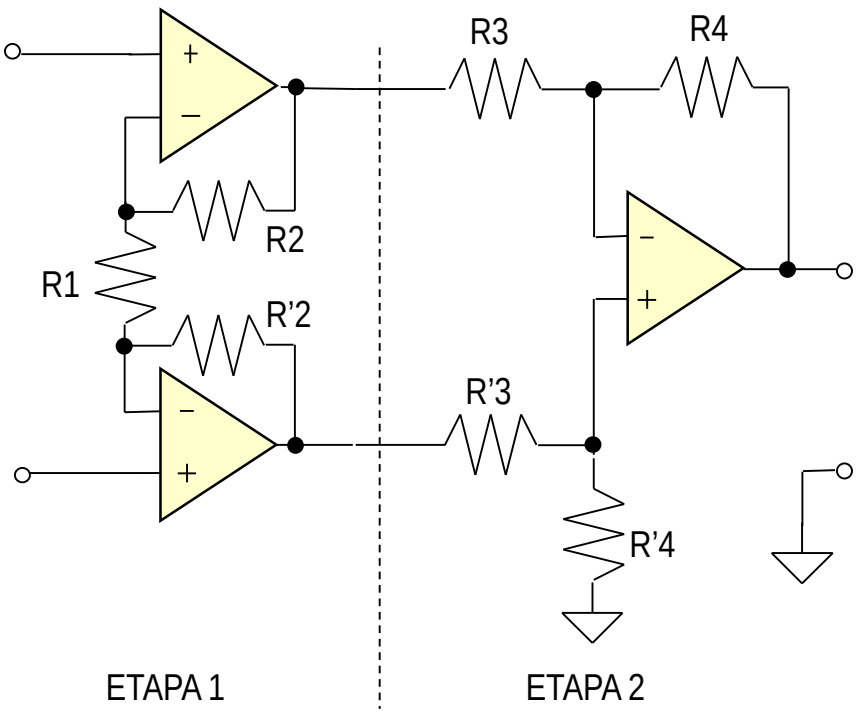
$$\vdots$$

$$CMRR_n = D_1 \cdot D_2 \cdot \ldots \cdot D_{n-1} \cdot C_n$$

$$\frac{1}{CMRR_T} \cong \frac{1}{CMRR_1} + \frac{1}{CMRR_2} + \cdots + \frac{1}{CMRR_n}$$

Conclusión:
Conviene poner toda la ganancia en las primeras etapas!!!

6) Amplificador de Instrumentación de 3 AO

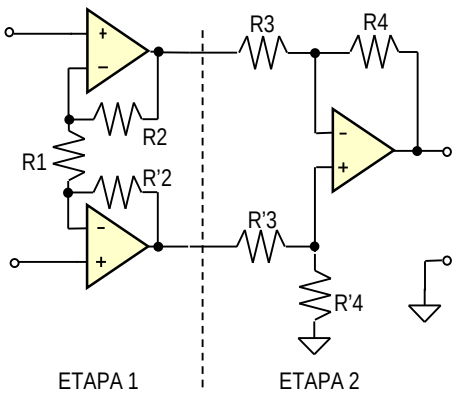


$$G_{dd} = ??$$

$$G_{dd} = G_{dd1} G_{dd2} = (1 + 2\frac{R_2}{R_1}) G_{dd2}$$

6) Amplificador de Instrumentación de 3 AO

Analizando el CMRR:



$$CMRR_1^{-1} = C_1^{-1} = \left(C_{AO2}^{-1} - C_{AO1}^{-1}\right) + \left(A_1^{-1} - A_2^{-1}\right)$$

$$CMRR_2^{-1} = \left(D_1 \cdot C_2\right)^{-1} = \left(\frac{1}{G_{DD1}}\right) \cdot \left(C_{AO3}^{-1} + \frac{4 \cdot t}{G_{DD2}}\right)$$

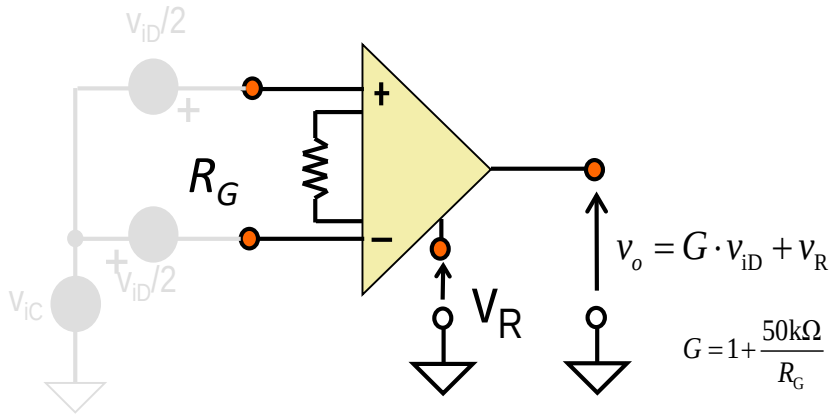
$$CMRR_T^{-1} = CMRR_1^{-1} + CMRR_2^{-1}$$

$$CMRR_T^{-1} = \left(C_{AO2}^{-1} - C_{AO1}^{-1}\right) + \left(A_1^{-1} - A_2^{-1}\right) + \frac{C_{AO3}^{-1}}{G_{DD1}} + \frac{4 \cdot t}{G_{DD1} G_{DD2}}$$

Observar que se verifica lo analizado previamente sobre cómo conviene repartir la ganancia!!!

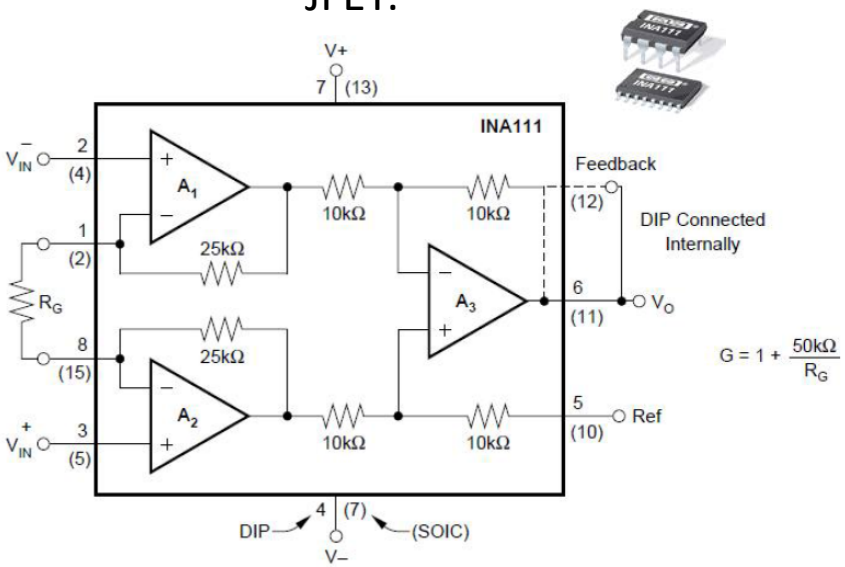
6) Amplificador de Instrumentación de 3 AO

Simbolización típica:

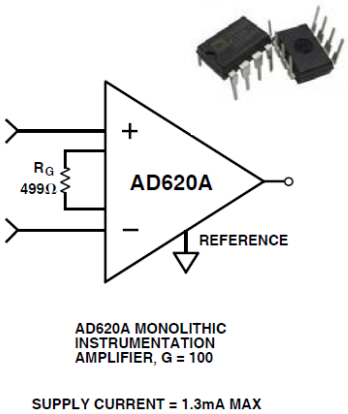


Presentación comercial en un único integrado:

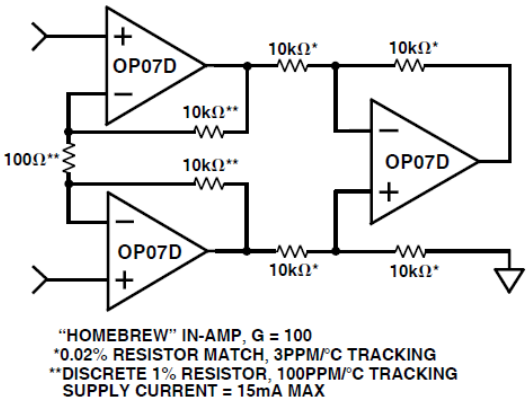
JFET:



BIPOLAR:



"CASERO":

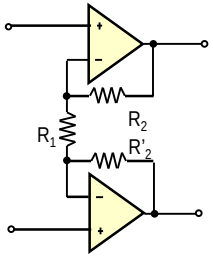
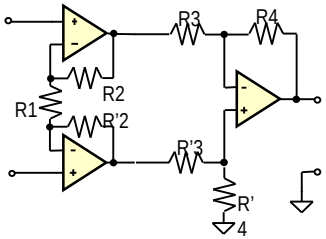
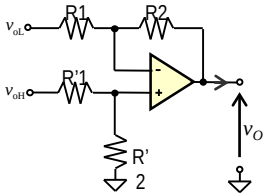
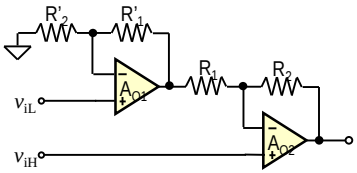


6) Amplificador de Instrumentación de 3 AO

Algunas Conclusiones:

- Adaptación de terminales
- Alta Zin.
- Alto CMRR.
- Solución existente en integrado comercial!

7) Cuadro Comparativo

	CMRR
	$\Delta C_{AO}^{-1} + \Delta A^{-1}$
	$\Delta C_{AO}^{-1} + \Delta A^{-1} + \frac{C_{AO3}^{-1}}{G_{DD1}} + \frac{4 \cdot t}{G_{DD1} G_{DD2}}$
	$C_{AO}^{-1} + \frac{4t}{G_{DD}}$
	$\Delta C_{AO}^{-1} + A_1^{-1} + \frac{4t}{G_{DD}}$

Fin